Asymmetric Hurwitz Extension of $\mathrm{GL}(2,\mathbb{C})$ Chiral Gravity

Clifford, Hurwitz, and the Arithmetic of Spinor Densities

L. McCulloch-James*

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Abstract

We propose an *asymmetric* extension of the $\mathrm{GL}(2,\mathbb{C})$ chiral first-order formalism for gravity in which the gravitational summand $\mathfrak{sl}(2,\mathbb{C})$ is left intact and only the abelian trace summand $\mathbb{C}$ is enlarged to the Hurwitz quaternion order $\mathbb{H}_{\mathrm{H}}$. The structural content of the move resides in the replacement of the complex determinant $\det \in \mathbb{C}^\times$ by the reduced norm $\mathrm{Nrd} \in \mathbb{R}_{>0}$, accompanied by a split of the ambient $\mathrm{SU}(2)$ phase data into a continuous magnitude and a discrete unit element of the binary tetrahedral group $2T = \mathbb{H}_{\mathrm{H}}^\times$. The complex density weight of $\mathrm{GL}(2,\mathbb{C})$ thereby refines into a real density weight together with a discrete internal sector. The representation theory of $2T$ has dimensions $(1,1,1,2,2,2,3)$ saturating $\sum d_i^2 = 24$, which we identify with the discrete skeleton of the Standard Model fermion sector: three generations from $2T/Q_8 \cong \mathbb{Z}_3$, electroweak doublets from the subgroup chain through V_4, and charge conjugation from $Z(2T) = \mathbb{Z}_2$. The soldering bivector Σ^{AB}, already present as the metric-defining object of the chiral formalism, plays the structural role of the Higgs vacuum expectation value: its on-shell value selects a complex sub-line $\mathbb{C}_\Sigma \subset \mathbb{H}$ and breaks $2T$ through the chain $2T \to Q_8 \to \mathbb{Z}_4 \to \mathbb{Z}_2$. We note in passing that the same asymmetric extension induces a mixed-order BGG complex on the linearised covariant phase space, but defer its full development to a sequel. We compare the construction with the Yang–Mills extension of Krasnov–Capovilla–Dell–Jacobson and with the F_4/24-cell programme of Farag-Ali *et al.*, and identify the calculations whose outcome would distinguish them.

Keywords: chiral gravity, Plebański formalism, Hurwitz quaternions, reduced norm, binary tetrahedral group, spinor densities, electroweak symmetry breaking.

Prerequisites. Spinor formalism for general relativity [31]; chiral / Plebański formulations of gravity [18, 32]; representation theory of $\mathrm{SL}(2,\mathbb{C})$ and finite quaternion groups [8, 34]. The interactive learner [22] provides the arithmetic background at greater leisure than the present paper allows.

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*Department of Natural & Unnatural Sciences, Dwight School London. `ljames@dwightlondon.org`.

1 Setting and Motivation

1.1 The chiral $\mathrm{GL}(2,\mathbb{C})$ formalism as Clifford rotors

The chiral first-order formulation of general relativity [6, 18, 32] takes as basic data a $\mathfrak{gl}(2,\mathbb{C})$-valued connection $\omega^A{}_B$ and a self-dual two-form Σ^{AB} valued in $\mathrm{Sym}^2(\mathbb{C}^2)$. The metric is recovered from Σ via the Urbantke formula [33]; the Hamiltonian sibling is the Ashtekar variables [2].

A point worth recording before saying anything about the structure of $\mathfrak{gl}(2,\mathbb{C})$: the gauge group $\mathrm{GL}(2,\mathbb{C})$ is not an arbitrary choice. The Clifford algebra Cl_3 of physical 3-space is sufficient to describe physical 4-spacetime, because the self-adjoint subspace of its universal covering group is precisely

$$\mathrm{Cl}_3^* \;=\; \mathrm{GL}(2,\mathbb{C}) \;=\; \left\{\, \xi \in \mathrm{GL}(2,\mathbb{C}) \,\middle|\, \xi^{AB} = f L^{AB},\ f^2 = \det(\xi^{AB}) \,\right\}, \tag{1}$$

the 8-dimensional real Lie group of *rotors* [26, 27]. A $\mathrm{GL}(2,\mathbb{C})$ transformation decomposes as $\xi^A{}_B = f\, L^A{}_B$, with $L \in \mathrm{SL}(2,\mathbb{C})$ and $f \in \mathbb{C}^\times$ satisfying $f^2 = \det(\xi)$, in the same way on dual and inverse indices:

$$\xi^A{}_B = f L^A{}_B, \qquad (\xi^{-1})^B{}_A = f^{-1}(L^{-1})^B{}_A. \tag{2}$$

Equivalently $\mathrm{GL}(2,\mathbb{C}) = \mathrm{SL}(2,\mathbb{C}) \otimes \mathbb{C}$ at the level of multiplicative structure. The factor f is the carrier of the complex density weight, magnitude and phase together; the factor L carries the chiral spinor transformation.

This decomposition organises a chain of natural subgroups,

$$\mathrm{GL}(2,\mathbb{C}) \;\supset\; \mathrm{UL}(2,\mathbb{C}) \;\supset\; \mathrm{SL}(2,\mathbb{C}), \qquad \dim_{\mathbb{R}} \;=\; 8 \supset 7 \supset 6, \tag{3}$$

where $\mathrm{UL}(2,\mathbb{C})$ is the 7-dimensional *unimodular* group of Lorentz dilations — the product, in any order, of a restricted Lorentz transformation and a homothety. $\mathrm{UL}(2,\mathbb{C})$ is the locus $|\xi| = 1$ at which weights and anti-weights overlap; $\mathrm{SL}(2,\mathbb{C})$ is the further restriction $\xi = 1$ at which the spinor transformation acts without any density weight. The two extra dimensions in passing from SL to GL are the real homothety (in UL) and the residual $\mathrm{U}(1)$ phase (beyond it). The machinery of spinor-density valued objects on this group was developed in unpublished notes of Plebański; the salient features are collected in McCulloch-James [25], and the structure is used as the starting point for the boundary-charge construction of McCulloch-James [24].

1.2 The Lie-algebra decomposition and the structural question

At the Lie-algebra level, the chain (3) corresponds to the decomposition

$$\mathfrak{gl}(2,\mathbb{C}) \;=\; \mathfrak{sl}(2,\mathbb{C}) \;\oplus\; \mathbb{C}, \tag{4}$$

identifying the traceless part with the chiral half of the Lorentz connection and the trace one-form $a := \frac{1}{2}\,\omega^{A}{}_{A}$ with an abelian electromagnetic potential. Reading (2) infinitesimally, a is the generator of the $\mathbb{C}$ summand and is the carrier of the density-weight data f, magnitude and phase together. Spinors of $\mathrm{SL}(2,\mathbb{C})$ lift to spinor densities of $\mathrm{GL}(2,\mathbb{C})$ transforming as $\psi \mapsto (\det \xi)^{w} \xi \psi = f^{2w} L\psi$.

Two structural features of (4) drive the present paper. First, the abelian summand is geometrically inert in the sense that nothing in the chiral action distinguishes it beyond its $\mathrm{U}(1)$ gauge structure; the arithmetic of this summand is free to be chosen, and the standard reading $\mathbb{C}$ is one option among several. Second, the determinant $\det \xi \in \mathbb{C}^{\times}$ (equivalently the density-weight carrier f with $f^2 = \det \xi$) that controls the spinor-density transformation is a fundamentally commutative object: once we contemplate replacing $\mathbb{C}$ by a non-commutative arithmetic, the determinant must itself be replaced, and its replacement is not a formal substitution of one symbol for another but a genuine restructuring of the data it carried.

The proposal of this paper is to replace the abelian summand by the Hurwitz quaternion order $\mathbb{H}_{\mathrm{H}}$ [8, 16, 34], leaving the $\mathfrak{sl}(2,\mathbb{C})$ sector intact:

$$\mathfrak{gl}(2,\mathbb{C}) \;=\; \mathfrak{sl}(2,\mathbb{C}) \oplus \mathbb{C} \quad\longrightarrow\quad \mathfrak{sl}(2,\mathbb{C}) \oplus \mathbb{H}_{\mathrm{H}}. \tag{5}$$

The resulting structure carries a discrete $2T$-skeleton whose representation theory matches the fermion content of the Standard Model (§2); replaces the complex density-weight carrier f by a real magnitude together with a discrete element of $2T$ (§3); fits naturally into a three-layer picture that distinguishes the Hurwitz move from a Clifford-style linearisation, on the rotor algebra Cl_3^* that *is* the chiral gauge group (§4); and breaks dynamically through the soldering bivector Σ^{AB} in the pattern $2T \to Q_8 \to \mathbb{Z}_4 \to \mathbb{Z}_2$ (§5).

The construction is structural rather than dynamical. We do not derive masses, mixing angles, or coupling constants. We argue that the discrete skeleton observed in nature is already encoded in the chiral $\mathrm{GL}(2,\mathbb{C})$ formalism for gravity, provided the abelian summand is read arithmetically rather than as a featureless $\mathbb{C}$, and provided the determinant is replaced by the reduced norm and the associated unit-quaternion sector. The interactive learner [22] develops the arithmetic substrate in greater detail than this paper attempts; the covariant phase space programme [24] develops the $\mathrm{GL}(2,\mathbb{C})$-as-rotors framing that the present paper builds on.

2 The Hurwitz Skeleton

2.1 From Eisenstein to Hurwitz

The complex numbers $\mathbb{C}$ contain two distinguished imaginary quadratic rings of class number one with non-trivial unit groups [10]:

$$\begin{array}{lll}\text{Gaussian:} & \mathbb{Z}[i], & U(\mathbb{Z}[i]) = \{\pm 1, \pm i\} \cong \mathbb{Z}_4, \\ \text{Eisenstein:} & \mathbb{Z}[\omega], \quad \omega = e^{2\pi i/3}, & U(\mathbb{Z}[\omega]) = \{\pm 1, \pm\omega, \pm\omega^2\} \cong \mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3.\end{array}$$

The Eisenstein unit group is the only imaginary-quadratic option that introduces a $\mathbb{Z}_3$ beyond the universal $\mathbb{Z}_2$ of charge conjugation. If the trace sector of $\mathfrak{gl}(2,\mathbb{C})$ is read with Eisenstein arithmetic, fermions coupled to it acquire a discrete $\mathbb{Z}_3$ phase that distinguishes three otherwise identical species [23].

The Eisenstein observation explains the count three but not the electroweak doublet structure or the mass hierarchy. Both require the Hurwitz extension, in which the $\mathbb{Z}_3$ is internalised as an automorphism of an honest 3-dimensional quaternionic space rather than a planar rotation of $\mathbb{C}$.

2.2 The Hurwitz order and its unit group

Let $\mathbb{H} = \mathbb{R}\langle 1, i, j, k\rangle$ denote the Hamilton quaternions. The *Lipschitz order* $L = \mathbb{Z}\langle 1, i, j, k\rangle$ is an order[1] in $\mathbb{Q} \otimes \mathbb{H}$ but is not maximal. Adjoining[2] the half-integer point $\rho = \frac{1}{2}(1 + i + j + k)$ produces the *Hurwitz order* [34, §11]

$$\mathbb{H}_{\mathrm{H}} \;=\; L + \mathbb{Z}\rho, \tag{6}$$

which is the unique maximal order in $\mathbb{Q} \otimes \mathbb{H}$ up to conjugacy. As an additive group, $\mathbb{H}_{\mathrm{H}}$ is isomorphic to the F_4 root lattice [7]. Direct calculation gives $\rho^2 = \frac{1}{2}(-1 + i + j + k)$ and $\rho^3 = -1$, so ρ has order 6 and the embedded subgroup $\langle\rho\rangle$ is the Eisenstein unit group $\mathbb{Z}_6$, with $\langle\rho^2\rangle$ the generation $\mathbb{Z}_3$ now living inside non-commutative arithmetic.[3]

The unit group $\mathbb{H}_{\mathrm{H}}^\times$ has order 24 and consists of

$$\mathbb{H}_{\mathrm{H}}^\times \;=\; \underbrace{\{\pm 1, \pm i, \pm j, \pm k\}}_{Q_8} \cup \underbrace{\left\{\tfrac{1}{2}(\pm 1 \pm i \pm j \pm k)\right\}}_{16\ \text{Hurwitz-proper}}. \tag{7}$$

This is the *binary tetrahedral group* $2T$, isomorphic to $\mathrm{SL}(2, \mathbb{F}_3)$ [9, 35, nLab], fitting into the non-split extension

$$1 \to \mathbb{Z}_2 \to 2T \to A_4 \to 1. \tag{8}$$

2.3 Subgroup lattice and the semidirect structure

The non-trivial normal subgroups of $2T$ are the centre $Z(2T) = \{\pm 1\} \cong \mathbb{Z}_2$ and the quaternion subgroup $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ of index three [35]. Crucially:

$$2T \;\cong\; Q_8 \rtimes \mathbb{Z}_3, \tag{9}$$

with $\mathbb{Z}_3$ generated by ρ^2 acting on Q_8 by cyclic rotation $i \to j \to k \to i$. The arithmetic source of $|2T| = 24$ is therefore not a cyclotomic field extension but the semidirect product of the Lipschitz quaternionic core (Q_8, order 8) with the Eisenstein cyclic permutation ($\mathbb{Z}_3$). The Eisenstein subring sits inside $\mathbb{H}_{\mathrm{H}}$ via $\omega \mapsto \rho^2$, and the planar $\mathbb{Z}_3$ of $\mathbb{Z}[\omega]$ becomes the spatial $\mathbb{Z}_3$ cycling (i, j, k).

2.4 Representation theory of $2T$

The character table of $2T$ has seven conjugacy classes and seven complex irreducible representations of dimensions $(1, 1, 1, 2, 2, 2, 3)$, satisfying $\sum d_i^2 = 24$ [**?** nLab]. The structure is summarised in Table 1.

Observation 2.1. The dimensions $(1, 1, 1, 2, 2, 2, 3)$ of the irreducibles of $2T$ saturate the order constraint $\sum d_i^2 = 24$. The empirical identification with the Standard Model fermion content $3\times$(generation)$+3\times$(doublet sector)$+1\times$(adjoint / colour seed) leaves no representation unaccounted for.

3 What Replaces the Determinant

The structural content of the move (5) is most visible in what happens to the determinant. Equivalently, in the language of §1: what happens to the density-weight carrier f, with $f^2 = \det(\xi)$, when we replace $\mathbb{C}$ by a non-commutative arithmetic. The complex determinant $\det\xi$ has three properties that pull together a great deal of structure: multiplicativity, $\det(\xi\eta) = \det\xi \det\eta$;

[1] See this interactive for background on orders Orders
[2] See this interactive for background on Adjoining
[3] See this interactive for background on Lipschitz order as the natural quaternionic analogue of $\mathbb{Z}(i)$

Irrep	Dim	Restriction to $\langle\rho^2\rangle \cong \mathbb{Z}_3$	Construction
$\mathbf{1}_0$	1	trivial	trivial rep
$\mathbf{1}_\omega$	1	ω	factors through $2T \to 2T/Q_8 \cong \mathbb{Z}_3$
$\mathbf{1}_{\omega^2}$	1	ω^2	conjugate of $\mathbf{1}_\omega$
$\mathbf{2}_a$	2	$1 \oplus 1$	defining SU(2) embedding of $2T$
$\mathbf{2}_b$	2	$\omega \oplus \omega^2$	$\mathbf{2}_a \otimes \mathbf{1}_\omega$
$\mathbf{2}_c$	2	$\omega^2 \oplus \omega$	$\mathbf{2}_a \otimes \mathbf{1}_{\omega^2}$
$\mathbf{3}$	3	$1 \oplus \omega \oplus \omega^2$	conjugation on $\mathrm{Im}(\mathbb{H})$, lifted from A_4

Table 1: Irreducible representations of $2T$, characters under restriction to the cyclic subgroup $\langle\rho^2\rangle \cong \mathbb{Z}_3$, and explicit construction. The three one-dimensional irreps factor through the generation quotient $2T/Q_8$; the three two-dimensional irreps are interrelated by tensoring with the $\mathbb{Z}_3$ characters; the three-dimensional irrep is the standard rotation representation of the tetrahedral group A_4.

invertibility-detection, ξ invertible iff $\det \xi \neq 0$; and commutative-field-valuedness, $\det \xi \in \mathbb{C}$. The third is what makes $f = \sqrt{\det \xi}$ raisable to a power and useful for tracking density weights. In $\mathrm{SL}(2,\mathbb{C})$ the determinant is fixed at 1 (so $f = 1$) and spinors transform without any density-weight factor; in $\mathrm{UL}(2,\mathbb{C})$ the magnitude $|f|$ is fixed at 1 but the phase is free; in $\mathrm{GL}(2,\mathbb{C})$ both magnitude and phase of $f = re^{i\theta/2}$ vary, and spinors lift to spinor densities $\psi \mapsto f^{2w} L\psi$. The factor f packages *magnitude and phase together* in a single $\mathbb{C}^\times$-valued object.

3.1 Why the determinant breaks over $\mathbb{H}$

Over a quaternionic matrix ring $M_2(\mathbb{H})$, the naive formula $ad - bc$ is ambiguous because multiplication in $\mathbb{H}$ does not commute. There is no quaternionic-valued function on $M_2(\mathbb{H})$ satisfying all three of the properties above. The standard replacement is the *reduced norm* $\mathrm{Nrd}(A)$, defined via the embedding $\mathbb{H} \hookrightarrow M_2(\mathbb{C})$

$$a + bi + cj + dk \;\longmapsto\; \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}, \tag{10}$$

extended entrywise to $M_2(\mathbb{H}) \hookrightarrow M_4(\mathbb{C})$, and taking the ordinary complex determinant of the 4×4 image [34]:

$$\mathrm{Nrd}(A) \;=\; \det_{\mathbb{C}}\big(\Phi(A)\big) \;\in\; \mathbb{R}_{\geq 0}. \tag{11}$$

On a single quaternion $q = a + bi + cj + dk$, the reduced norm specialises to $\mathrm{Nrd}(q) = q\bar{q} = a^2 + b^2 + c^2 + d^2$. Multiplicativity and invertibility-detection are restored. But the output is now real, not complex.

3.2 The magnitude–phase split

The complex density-weight factor packed magnitude and phase into a single $\mathbb{C}^\times$-valued object $f = re^{i\theta/2}$. The reduced norm keeps only the magnitude-squared: $|f|^2 = r^2 = |\det \xi| = \mathrm{Nrd}$. Where does the phase go? It does not vanish; it is just no longer scalar. Every non-zero quaternion factors polarly as

$$q \;=\; |q| \cdot u, \qquad |q| = \sqrt{\mathrm{Nrd}(q)} \in \mathbb{R}_{>0}, \qquad u \in \mathrm{SU}(2), \quad \mathrm{Nrd}(u) = 1. \tag{12}$$

The magnitude $|q|$ is the direct quaternionic analogue of $|f|$: it is the real square root of the reduced norm, where on the complex side $|f|$ was the real square root of $|\det \xi|$. The unit factor

u inherits the role of the phase $e^{i\theta/2}$, except that u is a unit quaternion acting on internal indices by left-multiplication, and is generally *not* a scalar, being an internal SU(2) rotation. The single complex density carrier has split into two structurally different pieces: a real scaling factor, and a non-commutative internal symmetry.

Remark 3.1. Hurwitz enrichment of the trace sector gives *density weights plus discrete internal sectors*. The complex determinant carried magnitude and phase as one $\mathbb{C}^{\times}$-valued object. The quaternionic story splits these into a real magnitude (the reduced norm, continuous) and a finite group element (the unit factor, discrete after Hurwitz restriction). The split is the actual content of the move $\mathbb{C} \rightsquigarrow \mathbb{H}_{\mathrm{H}}$ on the trace sector.

3.3 The Hurwitz layer on top

Restricting to Hurwitz arithmetic, the unit factor u is forced to lie in the maximal-order unit group:

$$u \;\in\; \mathbb{H}_{\mathrm{H}}^{\times} \;=\; 2T, \qquad |2T| = 24. \tag{13}$$

So the enriched transformation a spinor density undergoes, in the Hurwitz-extended setting, is schematically

$$\psi \;\longmapsto\; \mathrm{Nrd}(A)^{w}\, \rho_{2T}(u)\, A\psi, \qquad u \in 2T, \tag{14}$$

where ρ_{2T} is some representation of $2T$ acting on the internal indices. The continuous density weight $\mathrm{Nrd}(A)^{w}$ is the surviving scalar piece; $\rho_{2T}(u)$ is a finite internal sector, not a density weight at all. Hurwitz units cannot be treated as ordinary scalar weights unless one restricts to the centre $Z(2T) = \{\pm 1\}$, in which case $\rho_{2T}(u) = \pm 1$ and the construction collapses back to a sign, which is the discrete data that survives at the end of the breaking chain (§5).

(14) is the precise mathematical object the present paper proposes. The transformation law of a spinor density in the asymmetrically-extended chiral formalism factors into a continuous real weight times a discrete element of $2T$ in some representation, both multiplying the standard $\mathfrak{sl}(2,\mathbb{C})$-action. Which representation of $2T$ is selected, on which fermion species, is the content of the Standard Model identification of Table 1.

3.4 Volumetric and barycentric constraints

The replacement of det by Nrd developed in §3.3 is a magnitude-level story: the continuous density weight survives, the phase data is split off into a discrete unit-quaternion sector. A complementary structural reading of the same move identifies a second kind of constraint that the Hurwitz refinement brings in, of a sort that the original $\mathbb{C}$ summand could not carry.

Recall first that $\det = 1$ on $\mathrm{GL}(2,\mathbb{C})$ corresponds, by $\exp(\mathrm{tr}(X)) = \det(\exp(X))$, to the trace-free condition $\mathrm{tr}(X) = 0$ on the Lie algebra $\mathfrak{gl}(2,\mathbb{C})$. This is the infinitesimal version of the volumetric constraint: the parallelogram spanned by the columns of the matrix has fixed area, equivalently the generator has vanishing trace as an operator on $\mathbb{C}^2$. The locus is the unit sphere $|\det| = 1$ at the Lie-group level, the codimension-one subspace $\mathfrak{sl}(2,\mathbb{C}) \subset \mathfrak{gl}(2,\mathbb{C})$ at the algebra level.

A different trace-free condition controls the Eisenstein sector. The generator ω of $\mathbb{Z}_3$ acts on $\mathbb{C}^3$ by cyclic permutation,

$$P \;=\; \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P^3 = I, \qquad \mathrm{tr}(P) = 0, \tag{15}$$

with eigenvalues $\{1, \omega, \omega^2\}$. The defining Eisenstein identity

$$1 + \omega + \omega^2 \;=\; 0 \tag{16}$$

is the statement $\mathrm{tr}(P) = 0$ at the eigenvalue level: the sum of the eigenvalues of P vanishes. So (16) is also a trace-free condition, of additive/barycentric character rather than multiplicative/volumetric character. Geometrically, the three Eisenstein roots $\{1, \omega, \omega^2\}$ are the unique three equal-magnitude unit vectors in $\mathbb{C}$ whose centroid is at the origin, and the triangle they span is the fixed locus of the constraint, in the way that the unit sphere is the fixed locus of $|\det| = 1$.

The two constraints live on different ambient spaces and constrain different objects. The $\mathfrak{sl}$ condition acts on a matrix in $\mathfrak{gl}(2, \mathbb{C})$; the Eisenstein condition acts on the eigenvalues of a finite-order operator. They are nevertheless the same kind of constraint in a precise sense: both impose a single-equation, codimension-one, trace-free restriction on their respective ambient space, with the result that one degree of freedom is removed and the residual configuration is fixed. The dual reading is:

$$\begin{aligned} &\text{volumetric:} && \det = 1,\ \mathrm{tr} = 0 && \text{(unit sphere, parallelogram of fixed area)} \\ &\text{barycentric:} && 1 + \omega + \omega^2 = 0 && \text{(triangle, three unit vectors of zero centroid).} \end{aligned} \tag{17}$$

The triangle plays the role for the Eisenstein geometry that the unit sphere plays for the spinorial geometry. The original $\mathbb{C}$ summand of (4) cannot carry the barycentric structure, because $\mathbb{C}$ has no canonical cyclic-three configuration of unit vectors. The Hurwitz refinement $\mathbb{H}_{\mathrm{H}}$ carries the barycentric structure for free, because the cyclic generator ρ^2 lies inside $\mathbb{H}_{\mathrm{H}}$ with $1 + \rho^2 + \rho^4 = 0$ playing the role of (16) in the non-commutative setting.

A corollary worth noting: the entire 24-element unit configuration $2T \subset \mathbb{H}_{\mathrm{H}}^\times$ is barycentric. The eight Lipschitz units $\{\pm 1, \pm i, \pm j, \pm k\}$ pair into antipodes and sum to zero; the sixteen half-integer units pair similarly. So $\sum_{u \in 2T} u = 0$ in $\mathbb{H}$, and the centroid of the 24-cell vertex set is the origin. The Hurwitz refinement does not just import the Eisenstein barycentric condition through the embedded $\mathbb{Z}_3$; it propagates it to the full unit group.

Remark 3.2 (The structural payoff)**.** The asymmetric extension (5) is, on this reading, the unique way to make the chiral $\mathfrak{gl}(2, \mathbb{C})$ formalism carry both kinds of trace-free condition at once. The $\mathfrak{sl}(2, \mathbb{C})$ summand retains the volumetric constraint that defines the spinor sector. The $\mathbb{H}_{\mathrm{H}}$-extended trace summand acquires a barycentric constraint that the original abelian $\mathbb{C}$ could not support. The two constraints sit on orthogonal pieces of the algebra and do not interfere, but together they make the trace-free structure of the full extended algebra genuinely two-fold. This is one mathematical reading of why $\mathbb{H}_{\mathrm{H}}$ rather than some other arithmetic extension is forced: it is the unique maximal order whose unit group contains a non-trivial cyclic permutation with vanishing trace on $\mathrm{Im}(\mathbb{H})$.

4 Three Layers: Geometry, Clifford, Hurwitz

The construction of §3 involves three logically distinct operations on quadratic forms and lattices that are easy to conflate. A reader coming from spinor calculus may suspect that the Hurwitz extension is a kind of Clifford-style linearisation, since spinors and Pauli matrices operate in territory that looks superficially similar. The Hurwitz move is structurally different from Clifford's, and the distinction is worth recording explicitly.

Geometry. The starting datum is a real vector space equipped with a quadratic form g. Examples: $\mathbb{R}^n$ with the Euclidean metric; Minkowski space with the Lorentzian metric; $\mathbb{H} \cong \mathbb{R}^4$ with the norm $N(q) = a^2 + b^2 + c^2 + d^2$. At this layer one has a vector space and a quadratic invariant. Symmetry is the orthogonal group $\mathrm{O}(g)$ that preserves g. No vectors are multiplied together.

Clifford. Clifford's construction takes the quadratic form g and produces an associative algebra $\mathrm{Cl}(g)$ in which vectors satisfy $v^2 = g(v, v)$, equivalently $vw + wv = 2g(v, w)$. Reflections and rotations become products of vectors inside $\mathrm{Cl}(g)$; the spin group sits inside $\mathrm{Cl}(g)$ as a subset of the multiplicative group; spinors arise as modules over $\mathrm{Cl}(g)$. Pauli matrices represent Cl_3, Dirac matrices $\mathrm{Cl}_{1,3}$. The Clifford move is to package the quadratic form into an algebra and let the algebra act — it introduces multiplication where there was none.

Hurwitz. The Hurwitz move is structurally different. Quaternions already satisfy $q\bar{q} = a^2 + b^2 + c^2 + d^2$ as a built-in feature of multiplication, and the chiral $\mathrm{GL}(2, \mathbb{C})$ formalism for gravity is precisely the rotor subspace $\mathrm{Cl}_3^* = \mathrm{GL}(2, \mathbb{C})$ of the universal cover of the Clifford group, as recorded in equation (1). What remains, on the rotor algebra already carrying its quadratic form and its multiplicative structure, is to ask: which lattice sits cleanly inside it? The answer is the Hurwitz order $\mathbb{H}_{\mathrm{H}}$ with 24 unit-norm points, maximal among orders in $\mathbb{Q} \otimes \mathbb{H}$. This is not a linearisation, an algebra construction, or anything resembling Clifford's move. It is a refinement of the arithmetic data inside an algebra that already carries all the Clifford structure the chiral formalism uses. The unit group $2T$ that emerges from the refinement is a finite symmetry of the lattice, being discrete, not continuous, and not realised through products of vectors. The asymmetric extension (5) acts at this third layer alone, on the trace sector of an algebra (Cl_3^*) that already provides the second.

Layer	Operation	What is added
Geometry	Quadratic form g on a vector space	Invariant; orthogonal group; no algebra
Clifford	Encode g via $v^2 = g(v, v)$	Multiplication; spin group; spinors
Hurwitz	Maximal lattice in an algebra carrying g	Arithmetic; finite unit group $2T$

Table 2: Three distinct operations on quadratic forms and lattices. The present construction works at the Hurwitz layer, on an algebra where the Clifford layer is already realised by quaternionic multiplication.

The disambiguation matters because popular accounts of "octonions and the Standard Model" or "Clifford algebra unification" can blur the distinction between the Clifford layer (which gives spinors) and the arithmetic layer (which gives discrete symmetries from unit groups of orders). Here we operate at the arithmetic layer alone, on top of the algebra Cl_3^* that the chiral formalism already uses for its gauge group.

5 Symmetry Breaking via the Soldering Bivector

5.1 Two extensions, one chosen

The extension (5) is asymmetric in a precise sense. The symmetric extension $\mathfrak{gl}(2, \mathbb{C}) \to \mathfrak{gl}(2, \mathbb{H})$, in which both summands are quaternionified, is mathematically natural but physically untenable: a quaternionic Lorentz sector would predict gravitational degrees of freedom and spin-statistics modifications not observed. The asymmetric reading

$$\mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{C} \;\longrightarrow\; \mathfrak{sl}(2, \mathbb{C}) \oplus \mathbb{H}_{\mathrm{H}} \tag{18}$$

leaves the gravitational sector alone and enlarges only the abelian internal sector which is, where the density-weight data of §3 lives.

5.2 The soldering bivector as order parameter

The soldering two-form Σ^{AB} defines the spacetime metric via the Urbantke formula [33]

$$g_{\mu\nu} \;\propto\; \Sigma^{AB}_{\mu\rho}\, \Sigma_{AB\,\nu}{}^{\rho}. \tag{19}$$

On the asymmetrically-extended algebra, Σ does double duty. Reading $\mathbb{H}_{\mathrm{H}} \subset \mathbb{H}$ as a four-dimensional real internal space, the soldering bivector, which is required on shell to define a non-degenerate metric selects a preferred imaginary direction in $\mathbb{H}$, equivalently an almost-complex structure J with $J^2 = -1$ on the internal space. This selection is the mechanism that breaks the discrete $2T$ symmetry through the chain

$$2T \xrightarrow{\Sigma \text{ selects } \mathbb{C}_\Sigma} Q_8 \xrightarrow{\text{orientation of } \mathbb{C}_\Sigma} \mathbb{Z}_4 \xrightarrow{\text{conjugation}} \mathbb{Z}_2. \tag{20}$$

Proposition 5.1 (Stabiliser structure)**.** *Let $\mathbb{C}_\Sigma = \mathbb{R}\cdot 1 + \mathbb{R}\cdot u \subset \mathbb{H}$ for $u \in \mathrm{Im}(\mathbb{H})$ a unit imaginary (e.g. $u = i$). The setwise stabiliser of $\mathbb{C}_\Sigma$ in $2T$ under conjugation $v \mapsto gvg^{-1}$ is Q_8. The pointwise stabiliser (fixing u itself) is $\{\pm 1, \pm u\} \cong \mathbb{Z}_4$.*

Sketch. Take $u = i$. The pointwise stabiliser is the centraliser $C_{2T}(i) = \{\pm 1, \pm i\} \cong \mathbb{Z}_4$. The setwise stabiliser additionally includes elements that send $i \to -i$ while preserving $\mathbb{C}_i$ as a set; the eight elements $\{\pm j, \pm k\}$ all act this way, doubling the count to $|Q_8| = 8$. □

Remark 5.2 (Why setwise)**.** The electroweak doublet structure requires the setwise stabiliser Q_8. The pointwise version $\mathbb{Z}_4$ would give an unbroken U(1) phase distinguishing generations, which is not observed. Charge conjugation flips the orientation of $\mathbb{C}_\Sigma$ but preserves it as a set, so charge conjugation is unbroken at the doublet stage.

The first arrow of (20) gives the three-generation residual $2T/Q_8 \cong \mathbb{Z}_3$; the second gives the electroweak doublet structure as the orientation-changing quotient $Q_8/\mathbb{Z}_4 \cong \mathbb{Z}_2$; the third is charge conjugation.

5.3 Σ as the discrete Higgs

In the Standard Model the Higgs vacuum expectation value $\langle \Phi \rangle$ is an external object whose dynamical role is to break $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\mathrm{em}}$. In the asymmetric Hurwitz extension, the soldering bivector Σ^{AB} plays the same structural role but is not external: it is the metric-defining object already present in the chiral formalism. The two are not independent objects awkwardly related; they are the same object viewed from the gravitational and the internal sides of (18).

This is the precise sense in which observed electroweak symmetry breaking is the *motivation* for the construction rather than an obstacle. The standard picture posits a symmetry, posits a Higgs sector to break it, and posits a vacuum expectation value to set the scale. The Hurwitz picture has the symmetry forced by arithmetic, the breaking forced by soldering, and the order parameter identified with an object already required for gravitational reasons.

6 Mixed-Order Structure on the Linearised Phase Space

The asymmetric extension (18) induces a representation-theoretically asymmetric structure on the linearised covariant phase space, in the following sense. The linearised torsion equation

$$D_{\omega_0}(\delta\Sigma^{AB}) + \delta\omega^{(A}{}_C \wedge \Sigma_0^{B)C} = 0 \tag{21}$$

is algebraic in $\delta\omega = \delta\Gamma + \delta a \cdot \mathbf{1}$, with $\delta\Gamma \in \mathfrak{sl}(2,\mathbb{C})$ and $\delta a \in \mathbb{H}_{\mathrm{H}}$. The traceless part $\delta\Gamma$ can be solved algebraically and substituted, producing a second-order operator on the surviving field $\delta\Sigma$:

$$\delta F^A{}_B\big|_{\mathfrak{sl}} \in J^2(\delta\Sigma). \tag{22}$$

The trace contribution $\delta a \cdot \mathbf{1}$ is unconstrained by (21) (it can be absorbed by a redefinition $\omega_0 \to \omega_0 + \delta a$, since Σ has conformal weight $+1$ under the trace), and survives into the linearised dynamics with its order unchanged. The schematic result is a mixed-order complex:

$$\boxed{\begin{array}{llllllll} \mathfrak{sl}(2,\mathbb{C}): & B_0^{\mathrm{grav}} & \xrightarrow{D^{(1)}} & B_1^{\mathrm{grav}} & \xrightarrow{D^{(2)}} & B_2^{\mathrm{grav}} & \xrightarrow{D^{(1)}} & \cdots \\ \mathbb{H}_{\mathrm{H}}: & B_0^{\mathbb{H}} & \xrightarrow{d^{(1)}} & B_1^{\mathbb{H}} & \xrightarrow{d^{(1)}} & B_2^{\mathbb{H}} & \xrightarrow{d^{(1)}} & \cdots \end{array}} \tag{23}$$

where the superscripts denote differential (jet) order. The structure is the chiral analogue of the elasticity / Riemann–Cartan complexes of finite element exterior calculus [1, 15], with the asymmetric trace extension responsible for the mixed jet order.

We mention this structure here because it is forced by the same asymmetric extension that produces the discrete skeleton, and because the existence of a structure-preserving discretisation with two independent mesh scales; with a gravitational J^2 sector and a discrete arithmetic J^1 sector being the most natural numerical realisation of the construction. A full development of the BGG / FEEC machinery in the chiral $\mathfrak{gl}(2, \mathbb{H}_{\mathrm{H}})$ setting is deferred to a sequel.

Remark 6.1 (Constraint vs. dynamical coupling)**.** The $\mathfrak{sl}(2, \mathbb{C})$ and $\mathbb{H}_{\mathrm{H}}$ sectors are independent at the constraint level: the torsion-free condition $D\Sigma = 0$ does not constrain the trace perturbation δa, and the elimination acts on $\delta\Gamma$ alone. The two sectors are coupled at the dynamical level: matter equations involve the Hodge star $\star$, which depends on the metric, which depends on Σ_0 via (19). This is the chiral analogue of Maxwell theory in curved spacetime being kinematically decoupled from the gravitational sector at the level of $d^2 = 0$, and dynamically coupled through the metric in the Hodge star.

7 Comparisons

7.1 Yang–Mills extension

The Yang–Mills extension of the chiral formalism due to Capovilla et al. [6], Robinson [28], developed in Krasnov [18], takes

$$\mathfrak{sl}(2, \mathbb{C}) \ \oplus \ \mathfrak{g}, \tag{24}$$

where $\mathfrak{g}$ is a chosen compact Lie algebra. The 2-form formulation absorbs the Yang–Mills dynamics into the same Σ-based variational principle as gravity. The asymmetric Hurwitz extension (18) differs from (24) in three respects, summarised in Table 3.

Feature	Hurwitz extension	Yang–Mills extension
Internal algebra	forced ($\mathbb{H}_{\mathrm{H}}^{\times} = 2T$)	chosen
Generation count	forced ($\lvert 2T/Q_8 \rvert = 3$)	input
EW breaking pattern	forced (subgroup lattice + Σ)	posited
Order parameter	Σ^{AB} (already present)	Higgs (independent)
Density structure	Nrd $+\rho_{2T}(u)$ (split)	unrelated to gauge structure
Colour $\mathrm{SU}(3)_c$	via D_4 triality (less forced)	built in

Table 3: Comparison of the asymmetric Hurwitz extension with the Yang–Mills extension of the chiral formalism.

The Hurwitz arithmetic supplies what Yang–Mills takes as input (generation count, EW breaking pattern, order-parameter identification, the density-weight refinement); the Yang–Mills extension supplies what Hurwitz recovers only through a less-forced embedding (colour SU(3)). A combined picture with Hurwitz arithmetic on the abelian summand and Yang–Mills $\mathrm{SU}(3)_c$ on top would predict more and posit less than either route alone.

7.2 The 24-cell / F_4 programme

The recent programme of Farag-Ali et al. [11, 12] takes the 24-cell, equivalently the F_4 root system, as the fundamental quantum of spacetime, with the Standard Model gauge group recovered as the intersection of two F_4 stabilisers, $(\mathrm{SU}(3) \times \mathrm{SU}(3))/\mathbb{Z}_3 \cap \mathrm{Spin}(9) = (\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.

The arithmetic substrate is shared with the present construction: the vertices of the 24-cell are the Hurwitz units, and the F_4 root lattice is the additive group of $\mathbb{H}_{\mathrm{H}}$ [4, 36, 37]. The physical interpretation differs sharply. In [12], the 24-cell is fundamental and gravity emergent. In the present construction the chiral $\mathrm{GL}(2, \mathbb{C})$ formalism for gravity is primary, and the Hurwitz arithmetic appears as the discrete skeleton of the abelian summand only. This being a structural refinement of the trace sector, not a wholesale replacement of spacetime by a discrete polytope.

7.3 Discrete flavour symmetry models

Phenomenological models of lepton mass and mixing using A_4 as a flavour symmetry, following Ma and Rajasekaran [29] and reviewed by Altarelli and Feruglio [3], King and Luhn [17], arrive at the same finite group through a different route. The empirical motivation is tribimaximal mixing [14] and its variants; the symmetry A_4 is imposed to reproduce the data. The spinorial-double-cover variant $T' \cong 2T$ has been studied by Frampton and Kephart [13].

The structural observation underlying the present paper is that the discrete-flavour-symmetry community arrived at $2T$ by working backward from neutrino mixing data, while the Hurwitz route arrives at $2T$ by working forward from the arithmetic of the $\mathfrak{gl}(2, \mathbb{C})$ gauge algebra (specifically: from the reduced-norm refinement of the determinant on the trace summand). Convergence on the same finite group from such different starting points is the strongest indirect evidence for the construction.

8 Open Problems and Conclusion

The asymmetric Hurwitz extension is, on present evidence, a structural observation rather than a candidate physical theory. The following calculations would establish or refute its physical content.

(P1) Lie completion. The continuous gauge group emerging from the discrete $2T$ arithmetic, after soldering, must be argued to be uniquely $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, not some other Lie group containing $2T$ as a discrete subgroup. The relevant uniqueness result is presumably a maximality theorem in the spirit of Borel and de Siebenthal [5], applied to the chiral-soldering data.

(P2) Mass hierarchy. The mass ratios $m_e : m_\mu : m_\tau$ should be derivable from the Clebsch–Gordan coefficients of $2T \supset \mathbb{Z}_3$ together with the cubic $2T$-invariant. The democratic mass matrix at $\mathbb{Z}_3$-symmetric order has eigenvalues $(a + 2b, a - b, a - b)$; the splitting of the degenerate pair must come from the Σ-induced breaking of $\mathbb{Z}_3$ through the $\mathbf{2}_b \otimes \mathbf{1}_\omega$ branching. If the resulting ratios approximate $1 : 200 : 3500$ the framework merits serious development.

(P3) PMNS matrix. The mismatch between the $\mathbb{Z}_3$ eigenbasis and the mass eigenbasis should reproduce the PMNS mixing pattern, refining the A_4 results of Altarelli and Feruglio [3]. Since $A_4 \cong 2T/Z(2T)$, the Hurwitz framework must at minimum recover the A_4 phenomenology; it should additionally predict the small but non-zero θ_{13} correction discovered at Daya Bay/RENO.

(P4) Colour and the D_4 triality. The triplet representation $\mathbf{3}$ of $2T$ acts on $\mathrm{Im}(\mathbb{H}) = \mathrm{span}(i, j, k)$ by conjugation, with the $\mathbb{Z}_3 \subset 2T$ cycling the three imaginary units. This is the discrete skeleton of an SU(3) acting on three colour charges. Whether this lifts to a continuous $\mathrm{SU}(3)_c$ via the D_4 Dynkin triality on the 24-cell vertices [4], or whether a hybrid Hurwitz + Yang–Mills construction is required, is open.

(P5) The representation ρ_{2T}. Equation (14) requires the selection, for each fermion species, of a specific representation of $2T$ acting on its internal indices. The identification with Table 1 is structurally suggestive but not derived. A first-principles selection rule which would presumably be constrained by the requirement that ρ_{2T} be compatible with the Σ-induced breaking of §5, would convert the structural observation into a falsifiable assignment.

(P6) Coupling to surface charges on dynamical horizons. The chiral covariant phase space programme of McCulloch-James [24] constructs $\mathfrak{gl}(2,\mathbb{C})$-valued boundary charges on dynamical horizon segments, with first-law balance between the $\mathfrak{sl}(2,\mathbb{C})$ (gravitational) and $\mathbb{C}$ (electromagnetic) sectors of $\mathrm{GL}(2,\mathbb{C}) = \mathrm{Cl}_3^*$. The asymmetric Hurwitz extension promotes the trace charge to an $\mathbb{H}_{\mathrm{H}}$-valued object, and the first law acquires contributions from the discrete arithmetic of $2T$. The resulting balance law is the physical observable that distinguishes the asymmetric extension from the standard Einstein–Maxwell phase space.

Summary. The asymmetric Hurwitz extension (18) of the chiral $\mathrm{GL}(2,\mathbb{C})$ formalism is, structurally, a refinement of the trace summand from $\mathbb{C}$ to a maximal $\mathbb{Z}$-order in $\mathbb{H}$. The move replaces the complex determinant by the reduced norm, splitting the density-weight data into a continuous real weight and a finite $2T$-element. The discrete element occupies the place that, in the Standard Model, is occupied by the fermion generation/flavour structure: the irreducibles of $2T$ have dimensions $(1,1,1,2,2,2,3)$ saturating $|2T| = 24$ with no representation unaccounted for. The soldering bivector Σ^{AB} plays the role of the Higgs vacuum expectation value, breaking $2T$ through the chain $2T \to Q_8 \to \mathbb{Z}_4 \to \mathbb{Z}_2$. None of this derives the Standard Model. What it does identify is a specific arithmetic structure, already implicit in the chiral $\mathfrak{gl}(2,\mathbb{C})$ formalism, whose breaking pattern matches the one observed. Whether this is deep or merely decorative depends on the calculations enumerated above, principally the mass ratios from $2T$ Clebsch–Gordan coefficients.

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